Paper Plan (remove before submission)

1. **Introduction** — Motivate transformation-elasticity cloaking; establish that the ideal BGM polar tensor is unrealisable by Cauchy materials; survey existing approaches (arithmetic-mean symmetrisation, polar lattices, orthotropic database selection) and their trade-offs; state the contribution: direct differentiable search over four orthotropic Cauchy moduli per cell ($C_{11}, C_{12}, C_{22}, C_{66}$) via a coordinate neural field through a fully differentiable JAX-FEM pipeline, followed by microstructure realisability.

2. **Problem setup and FEM baseline** — Half-space geometry and Rayleigh-wave excitation; BGM push-forward derivation; triangular carpet shear map; ideal polar tensor $\mathbf{c}_{\text{eff}}$ (10 components, broken minor symmetry); cell discretisation on 14×10 Cartesian grid with 4CM Cauchy design variables; JAX-FEM solver and PML; cloaking metrics ($\mathcal{D}_{\text{r}}$, $\mathcal{D}_{\text{out}}$, cloak ratio η).

3. **Neural-field design pipeline** — Coordinate MLP outputting ($C_{11}, C_{12}, C_{22}, C_{66}, \rho$); Fourier positional encoding; multiplicative-residual initialisation from transformation-elasticity starting point (TODO confirm init source); loss function: cloaking term + drift penalty; adjoint backpropagation through FEM.

4. **Single-frequency results** — Compare continuous ideal cloak vs. optimised 4CM (per-cell Adam and neural-field reparameterisation) on 14×10 grid; mesh refinement study of the ideal continuous cloak; wavefield and material heatmap visualisations; symmetrised-tensor (Chatzopoulos 2023) closed-form baseline comparison.

5. **Broadband multifrequency optimisation** — Single-frequency overfitting demonstration; min–max/mean band objective and annealing schedule; frequency sweep results (50×50 grid, TODO confirm); Floquet–Bloch dispersion confirming physical admissibility of the broadband optimum.

6. **Microstructure-constrained optimisation** — CA dataset generation and periodic-FEM homogenisation; dataset distribution; GMM

realisability prior and constrained loss; diagnostic experiment separating matching gap (∼5%) from pixel-vs-homogenised gap (∼25%, hidden anisotropy); displacement-field validation; microstructure gallery; broadband micro-constrained extension.

7. **Discussion** — Comparison with Wang (2022) and Chatzopoulos (2023); limitations (square cells, isotropic descriptor); future work (generative realisability manifold, Willis dynamic homogenisation, 3-D extension, polar/Cosserat microstructures to close the broken-minor-symmetry gap).

8. **Conclusion** — Summarise: four-moduli Cauchy search via neural field as an alternative to closed-form symmetrisation; broadband training; microstructure realisability gap identified for the first time in the elastodynamic regime.

Neural-field design of broadband Rayleigh-wave carpet cloaks under microstructure realisability constraints

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Abstract

Transformation elasticity provides exact material distributions for elastodynamic cloaks, but the required stiffness tensors generally violate the minor symmetries of classical Cauchy elasticity and are difficult to realize using conventional materials. Existing approaches restore these symmetries by modifying or symmetrizing the transformed tensor, producing only an approximate cloak. Here we formulate 2D Rayleigh-wave carpet-cloak design as a PDE-constrained optimization problem, using a coordinate-based neural network and a differentiable finite-element solver to optimize symmetric stiffness and density fields by minimizing wave-field distortion. We consider both single-frequency and broadband optimization, with the broadband model trained simultaneously over multiple frequencies.

Physical realizability is addressed using a database of homogenized microstructures through conditional diffusion, autoencoder latent-space optimization, and nearest-neighbour selection. FEM simulations show that the unconstrained design approaches ideal transformation-based cloaking, while the best realizable design retains approximately X% of this performance, providing a direct route from continuum optimization to microstructured Rayleigh-wave cloaks.

Keywords: elastic cloaking, Rayleigh waves, transformation elasticity, neural field, implicit neural representation, metamaterials

1. Introduction

Transformation elasticity is a systematic way to steer elastic waves around an object. A coordinate map $\mathbf{x} = \Xi(\mathbf{X})$ opens up room for the object to hide

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in, and rewriting the elastodynamic equations in the mapped coordinates prescribes a position-dependent material in a surrounding coating that guides waves past the object as if it were not there [1, 2]. It is the mechanical counterpart of transformation optics [3]. The "carpet" variant, which hides a defect on an otherwise flat free surface [4], is the configuration most accessible to experiment [5, 6].

The coating material follows from the Brun–Guenneau–Movchan (BGM) push-forward, $c^{\mathrm{eff}}_{ijkl} = J^{-1}\, C_{IjKl}\, F_{iI}\, F_{kK}$ and $\rho^{\mathrm{eff}} = \rho\, J^{-1}$, where $\mathbf{F} = \nabla_X \mathbf{x}$ is the transformation gradient and $J = \det \mathbf{F}$ [1, 2]. The resulting tensor keeps the major symmetry of classical elasticity but breaks the minor symmetries, so the ideal cloak is a polar (Cosserat) medium that lies outside the class of ordinary Cauchy materials [7, 8]. Turning this exact but non-realisable prescription into a cloak that can be built from common materials is the problem we address.

We focus on Rayleigh waves, which travel along the free surface of a half-space in 2D. Their exact cloak is the polar tensor described above, and approaches to realising it with ordinary materials fall into three families: symmetrising it into a genuine Cauchy material, at the price of only approximate cloaking [9, 2]; reproducing the broken minor symmetry with a purpose-built lattice whose ordinary spring-and-mass elements, endowed with distributed restoring torques, assemble into a macroscopically polar (Cosserat) medium [10]; or selecting orthotropic Cauchy cells from a precomputed database, as done for quasi-static mechanical cloaks [11]. For Rayleigh waves specifically, Chatzopoulos et al. [12] compared triangular and semi-circular carpet cloaks and symmetrised their tensors by the arithmetic mean, reporting the symmetrised quadratic semi-circular design as the best performing.

The triangular geometry is, however, especially attractive. Its transformation is a shear with a constant gradient in each half of the cloak, so the ideal tensor is homogeneous there – two mirror-image phases – and nonsingular; being frequency-independent, it cloaks exactly at every frequency. Our approach to find a design inside the Cauchy class of ordinary materials is a differentiable pipeline built on a JAX-FEM solver [13] that evaluates a wave-field cloaking objective, with gradients flowing end-to-end through the FEM adjoint. We first use it to search for the best single realisable material – the orthotropic Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ that most closely reproduce the ideal response, rather than fixing them by symmetrisation. To improve on it while staying realisable, we then discretise the cloak into a regular grid of cells and let a coordinate-based neural field assign each cell its own Cauchy moduli and density; this per-cell freedom compensates for the broken minor symmetry better than any single homogeneous material can. We optimise at a single frequency and then across a target band. This improves substantially on the symmetrised designs of Chatzopoulos et al. [12]. The neural field acts as an implicit material representation, of the kind used for scene reconstruction in computer vision [14, 15, 16]. Reparameterising a design field by a neural network and optimising it through a differentiable solver has been used extensively in structural topology optimisation, with convolutional [17] and coordinate-based, mesh-free networks [18, 19]. Those methods minimise static compliance over a scalar density field; here the network instead outputs an anisotropic stiffness tensor and density and is trained against a frequency-domain elastodynamic cloaking objective, so as to approximate the non-symmetric transformation-elasticity material.

We then take the design one step further, to an explicit realisation. Each cell is filled with a two-phase (solid/void) microstructure drawn from a database of homogenised unit cells [20]; the cells share a single common base material and differ only in their internal geometry, so the whole cloak can be manufactured from one constituent by varying the pattern from cell to cell. To our knowledge no previous study has demonstrated a Rayleigh-wave elastodynamic cloak realised this way from a common material: earlier work either stops at effective symmetrised tensors [12], or reproduces the exact polar tensor only in principle – with a special lattice, exact only for a circular cloak in the infinite-cell limit [10]. We report both the gains of the microstructured cloak and the fraction of the ideal performance that survives homogenisation.

The paper is organised as follows. Section 2 sets up the half-space problem, the BGM push-forward for the triangular carpet, the cell discretisation, and the JAX-FEM solver and metrics. Section 3 introduces the neural-field pipeline and reports single-frequency results against the ideal and symmetrised baselines. Section 5 extends the design to a broadband objective. Section 6 constrains the cloak to realisable microstructures and validates the single-material realisation. Section 8 discusses the results and their limitations.

2. Problem setup and FEM baseline

Figure 1 summarises the computational configuration. A time-harmonic vertical point force applied on the traction-free surface of an isotropic half-space (λ, μ, ρ) launches a Rayleigh wave that propagates horizontally towards a triangular surface notch. The rectangular domain of width $W = 12.5\,b$ and depth H is truncated by Perfectly Matched Layer (PML) absorbing regions on the two lateral sides and the bottom; the top surface remains traction-free. The inset (Fig. 1b) shows the cloak geometry in detail: the triangular defect of depth a and half-width c is enclosed by the outer cloak boundary at depth b, whose two inclined faces define the inner and outer triangle boundaries z_1 and z_2; the two mirror phases of the cloak (related by the sign change of the shear F_{21} across the centreline $X_1 = 0$) are shaded separately.

2.1. Governing equations

We consider a homogeneous isotropic half-space with material properties (λ, μ, ρ), where λ and μ are the Lamé coefficients and ρ the mass density, and spatial coordinates $\mathbf{X} = (X_1, X_2)$ for the reference domain, with the free surface on $X_2 = 0$. For in-plane surface waves, i.e. Rayleigh waves, propagating along the horizontal X_1 direction, the governing Navier elastodynamic equation the following form

$$\nabla_X \cdot \big(\mathbf{C} : \nabla_X \mathbf{U}\big) = \rho\,\mathbf{U}_{tt}, \tag{1}$$

where $\mathbf{C}$ is the isotropic fourth-order elasticity tensor, $\mathbf{U} = (U_1, U_2)$ is the displacement, and $\mathbf{U}_{tt}$ denotes its second time derivative. Under the assumption of plane-strain elasticity, the elastic tensor can be written in Voigt notation $\{1, 2, 6\} = \{11, 22, 12\}$ as

$$C_{IJ} = \begin{bmatrix} \lambda{+}2\mu & \lambda & 0 \\ \lambda & \lambda{+}2\mu & 0 \\ 0 & 0 & \mu \end{bmatrix}, \qquad I, J = 1, 2, 6. \tag{2}$$

We apply a pointwise invertible transformation Ξ that maps the reference configuration (virtual domain) $\mathbf{X} \in \Psi$ to the deformed region (physical domain) $\mathbf{x} = \Xi(\mathbf{X}) \in \psi$ and the remaining domain to itself. Given $\mathbf{x} = (x_1, x_2)$ the coordinates of the physical domain, the transformation gradient and its determinant are

$$\mathbf{F} = \nabla_X \mathbf{x} = \begin{pmatrix} \dfrac{\partial x_1}{\partial X_1} & \dfrac{\partial x_1}{\partial X_2} \\ \dfrac{\partial x_2}{\partial X_1} & \dfrac{\partial x_2}{\partial X_2} \end{pmatrix}, \qquad J = \det(\mathbf{F}). \tag{3}$$

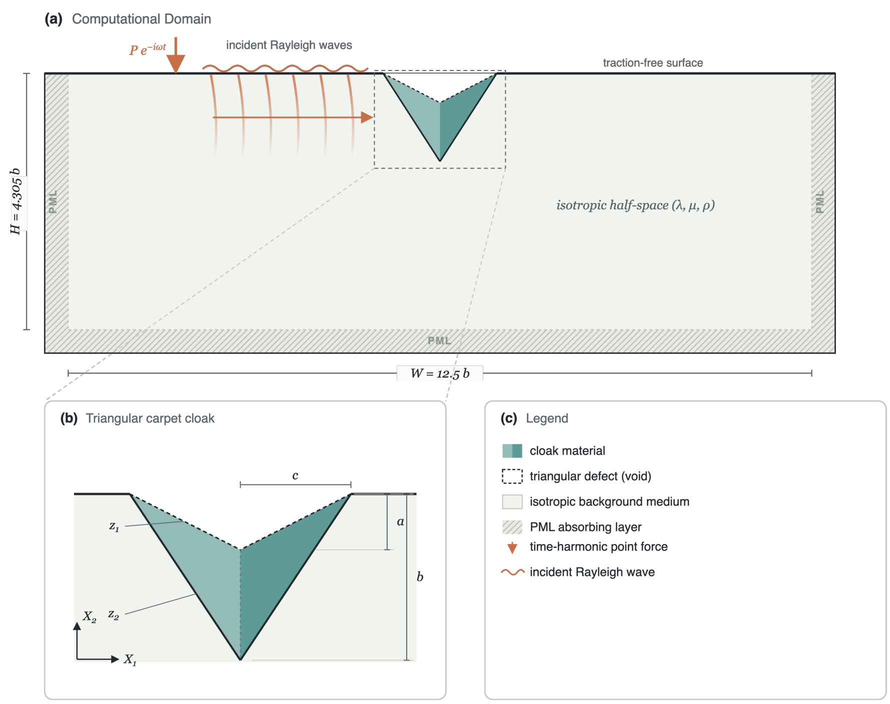


Figure 1: Problem configuration. **(a)** Computational domain: an isotropic elastic half-space (λ, μ, ρ) of width $W = 12.5\,b$ is truncated by PML absorbing layers on the lateral and bottom boundaries; the top surface is traction-free. A time-harmonic vertical point force (red arrow) applied upstream generates Rayleigh waves (orange wavefronts) that propagate towards the triangular cloak (teal). **(b)** Cloak geometry: the surface notch (void) has depth a and half-width c; the outer cloak boundary sits at depth $b = 3a$. Dashed lines mark the inner and outer triangle boundaries $z_1(X_1)$ and $z_2(X_1)$; the two mirror phases of the cloak are shaded in teal.

Equation (1) is not form-invariant under an arbitrary transformation Ξ [8]: the transformed material class depends on the gauge $\mathbf{U}(\Xi(\mathbf{X})) = \mathbf{A}\,\mathbf{u}(\mathbf{x})$ linking the virtual and physical displacements, with $\mathbf{A}$ a non-singular matrix. The choice $\mathbf{A} = \mathbf{F}$ leads to the Willis setting, which guarantees a symmetric stress tensor but requires two additional third-order coupling tensors and a tensorial mass density [8, 2, 21], physically replicable only within narrow frequency bands by resonant microstructures. For this reason we adopt the Cosserat setting $\mathbf{A} = \mathbf{I}$, following Norris and Shuvalov [2] and Brun et al. [1], for which the governing equation in the physical domain retains the Navier form

$$\nabla_x \cdot \big(\mathbf{c}^{\text{eff}} : \nabla_x \mathbf{u}\big) = \rho^{\text{eff}}\,\mathbf{u}_{tt}, \tag{4}$$

with the transformed mechanical parameters given by the Brun–Guenneau–Movchan (BGM) transformation relations

$$c^{\text{eff}}_{ijkl} = J^{-1}\,C_{IjKl}\,F_{iI}\,F_{kK}, \qquad \rho^{\text{eff}} = \rho\,J^{-1}. \tag{5}$$

The transformed tensor preserves the major symmetries ($c^{\text{eff}}_{ijkl} = c^{\text{eff}}_{klij}$) but not the minor ones,

$$c^{\text{eff}}_{ijkl} \neq c^{\text{eff}}_{jikl} \neq c^{\text{eff}}_{ijlk} \neq c^{\text{eff}}_{jilk}, \tag{6}$$

except for special cases such as conformal transformations. The medium is nonetheless described by a single fourth-order, non-symmetric and, eventually, inhomogeneous elastic tensor with scalar density: the stress $\sigma_{ij} = c^{\text{eff}}_{ijkl}\,\partial u_k/\partial x_l$ acts on the *full* displacement gradient – not the symmetric strain – and is itself non-symmetric (polar medium). All computations below are carried out in the frequency domain with the time-harmonic convention $\mathbf{u}(\mathbf{x}, t) = \mathrm{Re}\big[\mathbf{u}(\mathbf{x})\,e^{-\mathrm{i}\omega t}\big]$, which replaces $\mathbf{u}_{tt}$ by $-\omega^2\mathbf{u}$.

2.2. Transformation elasticity for the triangular carpet cloak

The cloak follows the pinched-carpet construction of Chatzopoulos et al. [12]. With the free surface on $X_2 = 0$ and the cloak centred at $X_1 = 0$, the inner (defect) and outer cloak boundaries are the triangles

$$z_1(X_1) = \frac{a}{c}\,|X_1| - a, \qquad z_2(X_1) = \frac{b}{c}\,|X_1| - b, \tag{7}$$

where a is the defect depth, b the cloak depth, and c the surface half-width. The shear map

$$x_1 = X_1, \qquad x_2 = \frac{z_2(X_1) - z_1(X_1)}{z_2(X_1)}\,X_2 + z_1(X_1) \tag{8}$$

compresses the triangular region above z_2 onto the cloak region between z_1 and z_2, opening the notch. Its gradient is piecewise constant,

$$\mathbf{F} = \begin{bmatrix} 1 & 0 \\ F_{21} & F_{22} \end{bmatrix}, \qquad F_{21} = \operatorname{sign}(X_1)\,\frac{a}{c}, \qquad F_{22} = \frac{b-a}{b} = J. \tag{9}$$

For an isotropic background (λ, μ) the BGM push-forward (5) then yields, in augmented Voigt ordering $[\sigma_{11}, \sigma_{22}, \sigma_{12}, \sigma_{21}]$, the effective stiffness

$$\mathbf{c}_{\text{eff}} = \begin{bmatrix} \dfrac{2\mu+\lambda}{F_{22}} & \lambda & 0 & \dfrac{F_{21}}{F_{22}}(2\mu+\lambda) \\ \lambda & \dfrac{F_{21}^2\mu + F_{22}^2(2\mu+\lambda)}{F_{22}} & \dfrac{F_{21}}{F_{22}}\,\mu & F_{21}(\lambda+\mu) \\ 0 & \dfrac{F_{21}}{F_{22}}\,\mu & \dfrac{\mu}{F_{22}} & \mu \\ \dfrac{F_{21}}{F_{22}}(2\mu+\lambda) & F_{21}(\lambda+\mu) & \mu & \dfrac{F_{21}^2(2\mu+\lambda) + F_{22}^2\mu}{F_{22}} \end{bmatrix}, \tag{10}$$

together with the constant effective density

$$\rho^{\text{eff}} = \frac{\rho}{J} = \frac{b}{b-a}\,\rho. \tag{11}$$

Unlike radial blow-up cloaks, whose tensors are singular at the inner boundary, the triangular tensor is non-singular and uniform within each half of the cloak (the shear F_{21} changes sign across the centreline $X_1 = 0$), so the ideal cloak consists of just two mirror-image material phases. The price is that *three* minor symmetry pairs are broken independently ($c_{11,12} \neq c_{11,21}$, $c_{22,12} \neq c_{22,21}$, $c_{12,12} \neq c_{21,21}$), and these are the entries that symmetrisation strategies average away by hand.

2.3. Realisable Cauchy parameterisation

Throughout this paper the designed cloak is parameterised by four free orthotropic Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ (4CM) and a scalar density, obtained by direct numerical search through the differentiable FEM. This class is dictated by the microstructures that must ultimately realise the cloak (Sec. 6): square two-phase unit cells invariant under the point group D_2, whose two edge-aligned mirror axes fix the form of the homogenised stiffness. Writing the plane-strain law in Voigt notation $\{1, 2, 6\} = \{11, 22, 12\}$, a general symmetric Cauchy tensor has six independent constants,

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{12}\end{bmatrix} = \begin{bmatrix}C_{11} & C_{12} & C_{16}\\ C_{12} & C_{22} & C_{26}\\ C_{16} & C_{26} & C_{66}\end{bmatrix} \begin{bmatrix}\varepsilon_{11}\\ \varepsilon_{22}\\ 2\varepsilon_{12}\end{bmatrix}. \tag{12}$$

The edge mirror $(x_1, x_2) \mapsto (x_1, -x_2)$ reverses the shear strain, so requiring invariance $\mathbf{C} = \mathbf{TCT}$ with $\mathbf{T} = \mathrm{diag}(1, 1, -1)$ forces $C_{16} = C_{26} = 0$ and reduces the stiffness to the block-diagonal orthotropic form

$$\mathbf{C}_{4\mathrm{CM}} = \begin{bmatrix}C_{11} & C_{12} & 0\\ C_{12} & C_{22} & 0\\ 0 & 0 & C_{66}\end{bmatrix}, \tag{13}$$

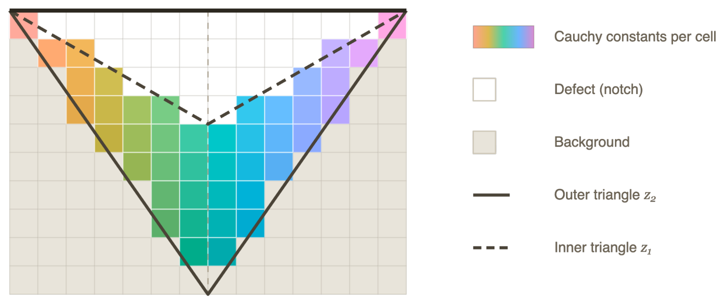
whose four moduli are exactly the per-cell design variables.

2.4. Cell discretisation

We discretise the cloak region into a regular Cartesian grid of $n_x \times n_y$ square cells (Fig. 2). Cells whose centre lies between the inner and outer triangles (7) are assigned a piecewise-constant effective stiffness and the density (11). Because the triangular transformation is affine in Cartesian coordinates, the tensor (10) is passed to the FEM solver without any rotation to a local frame. The ideal cloak tensor (10) is the full 4×4 augmented-Voigt polar tensor with ten independent components. Because real materials are Cauchy (minor-symmetric), the per-cell design variable is four free orthotropic Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ and a scalar density, which are searched numerically by the differentiable optimiser. A systematic investigation of grid resolution established 14×10 as the optimal discretisation, balancing approximation fidelity against per-iteration computational cost; this resolution is used in all single-frequency experiments reported below.

2.5. Simulation domain and FEM solver

We consider a rectangular computational domain of width $W = 12.5\,b$ and depth $H = 4.305\,b$, truncating the half-space. The top boundary is traction-free; PML absorbing layers terminate the two lateral boundaries and the bottom. A triangular surface notch of depth $a = 0.0774\,H$ and half-width $c = 0.1545\,H$ is centred on the free surface, coated by a cloak of depth $b = 3a$. The background medium is isotropic with $\rho = 1600\,\mathrm{kg/m^3}$, shear speed $c_s = 300\,\mathrm{m/s}$, and $c_p = \sqrt{3}\,c_s$ ($\nu = 1/4$, $\lambda = \mu = 144\,\mathrm{MPa}$,



14 × 10

Figure 2: Cartesian cell decomposition of the triangular cloak at the 14×10 resolution used in all single-frequency experiments. Coloured cells carry per-cell Cauchy constants $(C_{11}, C_{12}, C_{22}, C_{66}, \rho)$; white cells lie inside the defect (void notch); grey cells are background. The solid line is the outer cloak boundary z_2 and the dashed line is the inner boundary z_1.

plane strain). The excitation is a time-harmonic vertical point force applied on the free surface upstream of the cloak, generating Rayleigh waves that propagate along the surface towards the defect. Frequencies are reported in the dimensionless form $f^\star = f\,b/c_R$, where $c_R \approx 0.92\,c_s$ is the Rayleigh wave speed of the background: at $f^\star = 1$ the Rayleigh wavelength equals the cloak depth b. The nominal design frequency is $f^\star = 2$.

The frequency-domain elastodynamic equations are solved with complex-valued degrees of freedom and Rayleigh-damping PML through a JAX-FEM finite-element solver [13] on a triangular mesh with distance-based refinement near the cloak boundaries ($\sim$246,000 elements, $\sim$124,000 nodes). The entire pipeline from cell material parameters through assembly and sparse solve to the cloaking loss is differentiable via implicit adjoint differentiation through the linear FEM solve, enabling gradient-based optimisation of the per-cell stiffness and density.

2.6. Cloaking metrics

We use three relative L^2 metrics. The *right-boundary distortion*

$$\mathcal{D}_{\mathrm{r}} = 100\,\frac{\|\mathbf{u}_{\mathrm{cloak}} - \mathbf{u}_{\mathrm{ref}}\|_{L^2(\partial\Omega_{\mathrm{right}})}}{\|\mathbf{u}_{\mathrm{ref}}\|_{L^2(\partial\Omega_{\mathrm{right}})}}\,, \tag{14}$$

measures the transmission-side mismatch with respect to the defect-free reference. The *outside-cloak distortion* $\mathcal{D}_{\mathrm{out}}$ replaces the boundary integral by an integral over $\Omega_{\mathrm{phys}} \setminus \overline{\Omega}_{\mathrm{cloak}}$ and captures the global wavefield perturbation; it is closely related to the objective used by Wang et al. [11]. In broadband sections we also report the dimensionless cloak ratio $\langle|\mathbf{u}|\rangle/\langle|\mathbf{u}_{\mathrm{ref}}|\rangle$ over the free surface beyond the cloak footprint, following Chatzopoulos et al. [12]; this is the cloaking objective used in the microstructure-constrained pipeline.

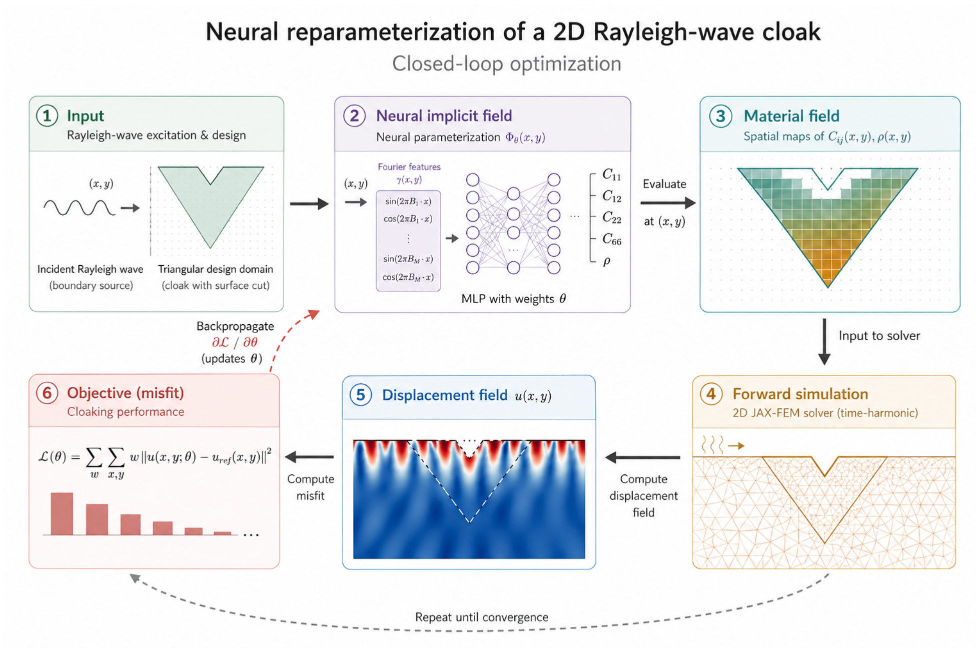


Figure 3: Neural-field design pipeline. Cell-centre coordinates are mapped through a Fourier positional encoding into a four-layer MLP that predicts the four free Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ and the density. The piecewise-constant material assignment is expanded to the FEM quadrature points and the cloaking objective is differentiated back through the implicit JAX-FEM adjoint into the network weights.

3. Neural-field design pipeline

3.1. Coordinate-conditioned material network

We reparameterise the cloak material field through a coordinate-based multilayer perceptron (MLP)

$$\Phi_\theta\colon (x, y) \;\mapsto\; (\mathbf{C}_{4\mathrm{CM}}(x, y),\; \rho(x, y)), \tag{15}$$

where θ denotes the network weights, $\mathbf{C}_{\mathrm{4CM}} \in \mathbb{R}^4$ contains the four free Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ — the design variables directly searched by the optimiser — and ρ is a positive scalar density. The network is evaluated at each cloak-cell centre to produce the piecewise-constant material assignment that is then expanded to the FEM quadrature points. A four-layer MLP with 256 hidden units ($\sim$200,000 weights) is used in all experiments; we adopt tanh activations and the Fourier positional encoding of Tancik et al. [16] so that the network can represent sharp material transitions near the cloak boundaries.

Figure 3 shows the architecture and the data flow through the differentiable FEM. Gradients $\partial\mathcal{L}/\partial\theta$ are obtained by backpropagating through the JAX-FEM adjoint solve and into the network.

3.2. Transformation-elasticity prior

The network does not learn the cloak from scratch. The four Cauchy moduli per cell are initialised at a starting point $\mathbf{C}^{(0)}_{\mathrm{4CM}}$ and then freely optimised. We tried two choices for this starting point – the arithmetic-mean symmetrisation of the ideal tensor (10) and the mean of the microstructure dataset (Sec. 6) – and found no statistically significant difference in the converged results, indicating that the design is driven by the cloaking objective rather than by the initialisation. Concretely, the last layer is scaled near zero ($\mathbf{W} \leftarrow 0.01\,\mathbf{W},\ \mathbf{b} \leftarrow \mathbf{0}$) and the network output is applied as a *multiplicative residual* with default output scale $\epsilon = 0.1$:

$$\mathbf{C}_{\mathrm{4CM}}(x,y) = \mathbf{C}^{(0)}_{\mathrm{4CM}}(x,y) \odot \left(1 + \epsilon\, \Phi^{(C)}_\theta(x,y)\right), \tag{16a}$$

$$\rho(x,y) = \rho^{(0)}(x,y)\left(1 + \epsilon\, \Phi^{(\rho)}_\theta(x,y)\right). \tag{16b}$$

The multiplicative form makes the relative perturbation spatially uniform regardless of the orders of magnitude separating the stiffness and density entries within each cloak phase. An additive variant is also implemented but produces more poorly conditioned gradients in practice.

3.3. Physical admissibility of the material output

The multiplicative residual (16) places no bound on the sign or the anisotropy of the moduli: an unconstrained MLP can drive a stiffness negative, break the positive-definiteness of the Cauchy tensor, or reach an anisotropy ratio no real microstructure can realise. Rather than police these violations with penalty terms after the fact, we build them into the decoding, so that *every* network output — at initialisation, along the optimisation path, and at convergence — is a physically admissible material by construction. The optimiser therefore searches only over the manifold of valid tensors and never has to be projected back onto it.

The network still emits four unbounded raw channels $\mathbf{r} = (r_{11}, r_{22}, r_{66}, r_{12}) = \Phi_\theta^{(C)}(x, y)$ per cloak cell, but these are mapped to the moduli through three guarantees.

(i) Positive diagonal and shear. The three positive-definite entries are set by a log-space (hence strictly positive) multiplicative residual,

$$C_{ii} = C_{ii}^{(0)} \exp(\epsilon\, r_{ii}), \qquad ii \in \{11, 22, 66\}, \tag{17}$$

which keeps them positive for any real r_{ii} and recovers $C_{ii}^{(0)}$ at $\mathbf{r} = \mathbf{0}$.

(ii) Bounded anisotropy. To exclude ratios beyond what the microstructure can attain, the diagonal ratio C_{11}/C_{22} is confined to $[1/R,\, R]$ while its geometric mean $\sqrt{C_{11}C_{22}}$ is held fixed. Writing $g = \frac{1}{2}\log(C_{11}/C_{22})$, a tanh squash on the half-log ratio,

$$g_b = \tfrac{1}{2}\log R\, \tanh\!\Big(\frac{g}{\frac{1}{2}\log R}\Big), \qquad C_{11}, C_{22} = \sqrt{C_{11}C_{22}}\, e^{\pm g_b}, \tag{18}$$

bounds the ratio smoothly to $[1/R, R]$ with R the anisotropy cap (we use $R = 30$, chosen above the 99th percentile of the microstructure dataset with margin).

(iii) Positive-definite coupling. The off-diagonal C_{12} is written through a correlation coefficient $\rho_c \in (-\kappa, \kappa)$ rather than directly,

$$C_{12} = \rho_c \sqrt{C_{11}C_{22}}, \qquad \rho_c = \kappa\, \tanh\!\big(r_{12} + \phi_0\big), \tag{19}$$

with $\kappa < 1$ (we use $\kappa = 0.99$). Because $|\rho_c| < \kappa < 1$, the orthotropic block stays strictly positive-definite, $\det = C_{11}C_{22}(1 - \rho_c^2) > 0$, for any r_{12}, and C_{12} is free to change sign. The offset $\phi_0 = \operatorname{arctanh}(\rho_c^{(0)}/\kappa)$ centres the map so that $\mathbf{r} = \mathbf{0}$ again recovers the initial coupling $C_{12}^{(0)}$. Non-cloak cells bypass the decoding and keep their background values exactly.

Together, (17)–(19) turn the raw network outputs into a smooth, everywhere-differentiable surjection onto the set of positive-definite, bounded-anisotropy Cauchy tensors, so the transformation-elasticity initialisation is recovered at $\mathbf{r} = \mathbf{0}$ and every gradient step lands on an admissible design.

3.4. Loss function

The optimisation minimises a *magnitude-band integral*: the mean-squared relative error of the displacement magnitude over a strip of depth $d = 0.5\,b$ (half the cloak depth) below the free surface, downstream of and excluding the cloak/defect footprint,

$$\mathcal{L}_{\text{cloak}}(\theta) = \frac{1}{|\mathcal{B}|}\int_{\mathcal{B}} \left(\frac{|\mathbf{u}_{\text{cloak}}|}{|\mathbf{u}_{\text{ref}}|} - 1\right)^2 \mathrm{d}A, \tag{20a}$$

$$\mathcal{B} = \left\{(x, y) \in \Omega_{\text{phys}} \setminus \overline{\Omega}_{\text{cloak}} : y_{\text{top}} - d \leq y \leq y_{\text{top}},\ x > x_c\right\}. \tag{20b}$$

where $\mathbf{u}_{\text{cloak}}$ and $\mathbf{u}_{\text{ref}}$ are the cloaked and defect-free reference fields and x_c is the downstream edge of the cloak. Comparing magnitudes rather than the complex fields themselves makes the objective phase-invariant, which is a deliberate advantage here: a wave that is diverted around the defect follows a slightly longer path and arrives with a small phase lag relative to the flat-surface reference. An explicit neighbour-smoothness term used in raw-parameter baselines [11] is unnecessary here: the neural field is spatially continuous by construction, and its smoothness is controlled directly through the Fourier positional encoding [16]. The number and bandwidth of the Fourier features set the spatial frequency content the network can represent—low frequencies yield smooth, gently varying material fields, whereas adding higher-frequency components lets the field resolve sharper transitions near the cloak boundaries.

4. Single-frequency results

All single-frequency results are reported at the design frequency $f^\star = 2.0$ through the dimensionless *cloak ratio*

$$\eta \;=\; \frac{\langle|\mathbf{u}|\rangle}{\langle|\mathbf{u}_{\text{ref}}|\rangle} \tag{21}$$

of the mean surface-displacement magnitude on the free surface beyond the cloak footprint to that of the defect-free reference (Sec. 2), so that $\eta \to 1$ is perfect cloaking and $|1-\eta|$ measures the residual distortion. We build up the design in three steps of increasing material freedom: a single homogeneous realisable material (Sec. 4.1), a handful of independent materials (Sec. 4.2), and finally a cell grid selected by a convergence sweep (Secs. 4.3–4.4).

4.1. A single realisable material already beats symmetrisation

We first ask how far a *single* realisable material can go. Both cloak phases are collapsed onto one orthotropic 4CM tile (a 1×1 cell filling the whole cloak) and its four Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ and density are searched by the differentiable FEM at $f^\star = 2.0$. The natural point of comparison is the closed-form arithmetic-mean symmetrisation of Chatzopoulos et al. [12]: the ideal polar tensor (10) made minor-symmetric by averaging each broken pair, instantiated on the same mesh and solved with the same FEM. Both are single homogeneous Cauchy materials, so this is a like-for-like test of hand symmetrisation against direct search.

Table 1 reports the outcome. The symmetrised tensor recovers only $\eta = 0.63$ and actually distorts the near field more than the bare notch (the ideal-material minor-symmetry violation is large at $f^\star = 2$). A single searched orthotropic material instead reaches $\eta = 0.85$ — a substantial jump towards the ideal continuous polar cloak ($\eta = 0.999$) obtained from just one degree of material freedom per phase, and clear evidence that direct search inside the Cauchy class outperforms symmetrisation. It is, however, still far from perfect: one material cannot reproduce the depth variation of the ideal tensor. Figure 4 confirms that the advantage is concentrated around the design frequency, as expected for a single-frequency optimisation.

Table 1: Single-frequency cloak ratio η (21) at $f^\star = 2.0$ for a single homogeneous realisable material. A single orthotropic 4CM tile found by direct search ($\eta = 0.85$) markedly outperforms the closed-form arithmetic-mean symmetrisation of Chatzopoulos et al. [12] ($\eta = 0.63$), but neither reaches the ideal continuous polar cloak. Values from `output/A_symmetrized` and `output/A_1x1flat4_vs_symmetrized`.

Configuration	η	$\lvert 1-\eta \rvert$
Uncoated notch (no cloak)	0.37	0.63
Symmetrised ideal (Chatzopoulos, closed-form Cauchy)	0.634	0.366
Single optimised material (1×1, 4CM)	0.850	0.150
Ideal continuous $\mathbf{c}_{\text{eff}}$ (polar, unrealisable)	0.999	0.001

4.2. Four materials reach 95% *cloaking*

Giving each phase more freedom closes most of the remaining gap. Splitting the cloak into a coarse $n_x \times n_y$ grid of independent orthotropic tiles and optimising all of them jointly, we find that even a 2×2 decomposition —

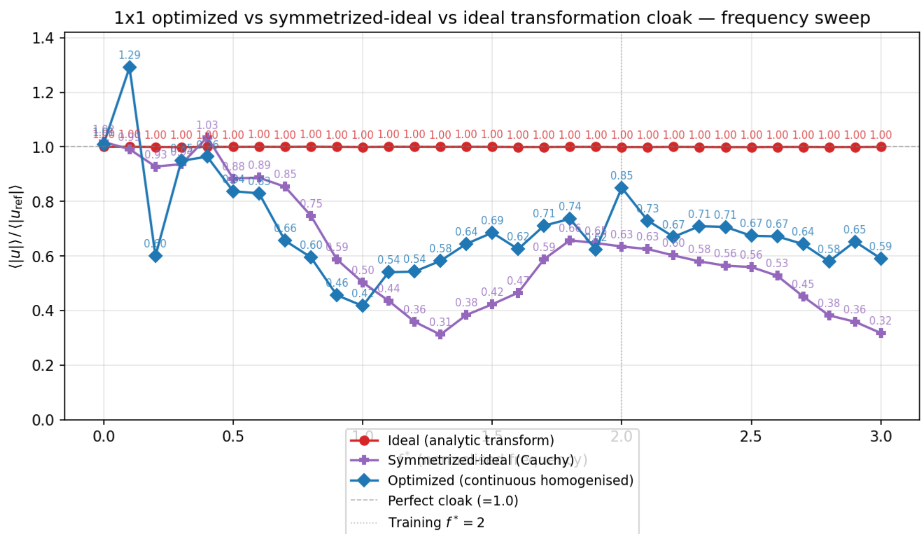


Figure 4: Frequency sweep of the cloak ratio $\eta = \langle|\mathbf{u}|\rangle/\langle|\mathbf{u}_{\text{ref}}|\rangle$ for the single-material designs: the ideal continuous transformation cloak (red, $\eta \approx 1$ at every frequency), the closed-form symmetrised tensor of Chatzopoulos et al. [12] (purple), and the single optimised 1×1 material (blue). At the training frequency $f^\star = 2$ the optimised material recovers $\eta = 0.85$ against 0.63 for symmetrisation.

just *four* realisable materials — lifts the cloak ratio to $\eta = 0.962$, i.e. within 4% of a perfect cloak (Table 2). The improvement is markedly anisotropic: subdividing in depth helps far more than in the propagation direction (1×2 reaches $\eta = 0.92$ whereas 2×1 actually drops to $\eta = 0.78$), reflecting that the ideal cloak material varies mainly with depth through the shear map (8). Beyond four materials the gain per added cell is small, motivating a more systematic grid study.

Table 2: Single-frequency cloak ratio η for small cell decompositions of the cloak (each cell is one independent orthotropic 4CM material, all optimised jointly at $f^\star = 2.0$). Four materials (2×2) already reach $\eta = 0.962$. Values from `output/A_partition_pushed`.

Decomposition $n_x \times n_y$	Materials	η	$\lvert 1 - \eta \rvert$
1×1	1	0.857	0.143
2×1	2	0.780	0.220
1×2	2	0.920	0.080
3×1	3	0.939	0.061
2×2	**4**	**0.962**	**0.038**
3×2	6	0.964	0.036

4.3. Grid selection by a cell-convergence sweep

To push towards perfect cloaking we refine the grid, but the resolution must be chosen so that cloaking has *converged* while the per-iteration cost stays low. We therefore run a single-frequency cell sweep over seven Cartesian grids from 5×4 to 32×24, optimising each to convergence and recording the cloak ratio (Table 3, Fig. 5).

A key point is that materials are assigned only to cells whose centre falls between the inner and outer triangles (7), so an $n_x \times n_y$ grid uses far fewer than $n_x n_y$ materials: the triangular cloak occupies only a fraction of its Cartesian bounding box. The 14×10 grid nominally spans 140 cells but places only **46 cells inside the cloak region**; these 46 are the actual design variables, the remaining 94 being background. (The figures in parentheses in Fig. 5 are these in-cloak counts: $5 \times 4 \to 7$, $8 \times 6 \to 16$, $10 \times 8 \to 28$, $14 \times 10 \to 46$, and so on.)

The cloak ratio saturates on a plateau at $\eta \approx 0.99$ ($\lvert 1-\eta \rvert \approx 0.01$) by 14×10: the three finer grids (20×15, 26×20, 32×24) give no further improvement — and even a marginal regression from FEM cost and optimisation noise —

while their cell counts, and hence solve cost, grow several-fold. The 14×10 grid is the smallest decomposition sitting firmly on this converged sub-1% plateau, and we adopt it for all subsequent single-frequency work: it is stable (on the flat part of the curve), cheap (only 46 active materials), and fast to optimise.

Table 3: Single-frequency cell-convergence sweep at $f^\star = 2.0$. Each grid is optimised to convergence; "cloak cells" is the number of cells whose centre lies inside the triangular cloak and therefore carries design variables (far fewer than $n_x n_y$). The cloak ratio plateaus at $\eta \approx 0.99$ by 14×10; finer grids do not improve it. Values from `output/A_cell_sweep`.

Grid $n_x \times n_y$	Total cells	Cloak cells	η	$\lvert 1-\eta \rvert$
5×4	20	7	0.944	0.056
8×6	48	16	0.943	0.057
10×8	80	28	0.977	0.023
14×10	**140**	**46**	**0.990**	**0.010**
20×15	300	100	0.990	0.010
26×20	520	176	0.988	0.012
32×24	768	256	0.988	0.012

4.4. *Near-perfect cloaking on the* 14×10 *grid*

The adopted 14×10 design (46 optimised in-cloak materials) reaches $\eta = 0.990$, effectively closing the gap to the ideal continuous cloak. Figure 6 compares the real-displacement magnitude of the optimised cloak against three references on the same mesh: the defect-free reference, the uncloaked notch, and the ideal continuous polar $\mathbf{c}_{\text{eff}}$. The uncloaked notch back-scatters the incident Rayleigh wave and casts a clear shadow downstream; the optimised 14×10 cloak restores the surface wavefield to the reference, matching the ideal continuous cloak panel while using only realisable orthotropic Cauchy materials. The only visible residual is a slight phase shift of the optimised solution: Figure 7 zooms into the cloak region and shows that, whereas the ideal continuous cloak keeps its near-surface wavefronts phase-aligned with the reference, the optimised 14×10 cloak carries them $\approx 0.18\lambda$ ($\approx 65°$) later. This lag is a direct and benign consequence of the phase-invariant magnitude loss (20): the objective constrains only the amplitude envelope, leaving a constant phase offset — the small extra path of a wave routed around the notch — unpenalised. Figure 8 shows the corresponding

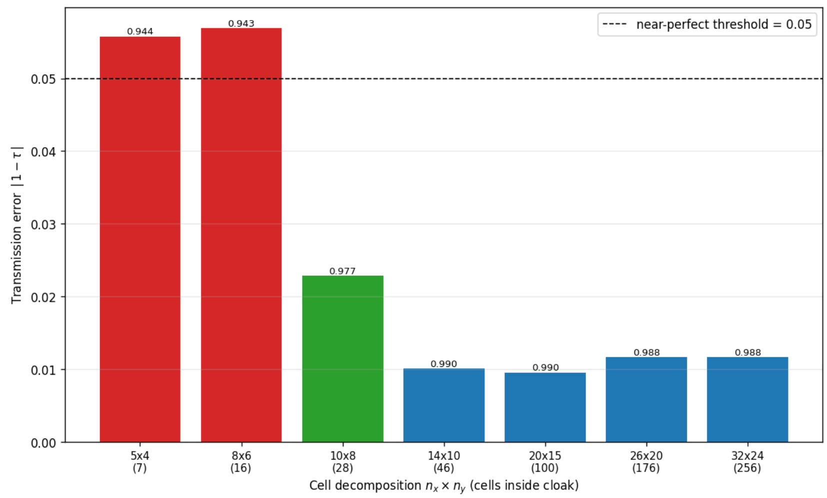


Figure 5: Cloaking error $|1 - \eta|$ versus cell decomposition in the single-frequency sweep; numbers in parentheses on the axis are the in-cloak cell counts. The error drops sharply up to 10×8 (28 cloak cells) and then flattens: 14×10 (46 cloak cells) is the smallest grid on the converged sub-1% plateau, with finer grids giving no further improvement.

optimised per-cell moduli $(C_{11}, C_{22}, C_{12}, C_{66})$ and density over the 46 cloak cells.

5. Broadband multifrequency optimisation

5.1. Single-frequency overfitting

A neural-field cloak trained at a single design frequency $f^\star$ overfits aggressively: the cloak loss exhibits a sharp minimum at $f^\star$ and degrades quickly away from it (Fig. 9). Two independent runs trained at $f^\star = 2.5$ and $f^\star = 3.0$ produce cloaks whose responses are nearly disjoint outside a band of width $\Delta f^\star \approx 0.2$ around their respective design frequencies. This behaviour is structurally identical to narrow-band overfitting in topology-optimised acoustic cloaks [22] and motivates a multifrequency training objective.

5.2. Reference sweeps

Before introducing the optimised broadband design we record two reference frequency sweeps on the chosen 50×50 cell decomposition: the ideal continuous $\mathbf{c}_{\text{eff}}$ cloak and the uncoated defect ("obstacle"). Both are evaluated on the same N-point grid $\{f_1, \ldots, f_N\}$ spanning the target band $[f_{\min}, f_{\max}]$.

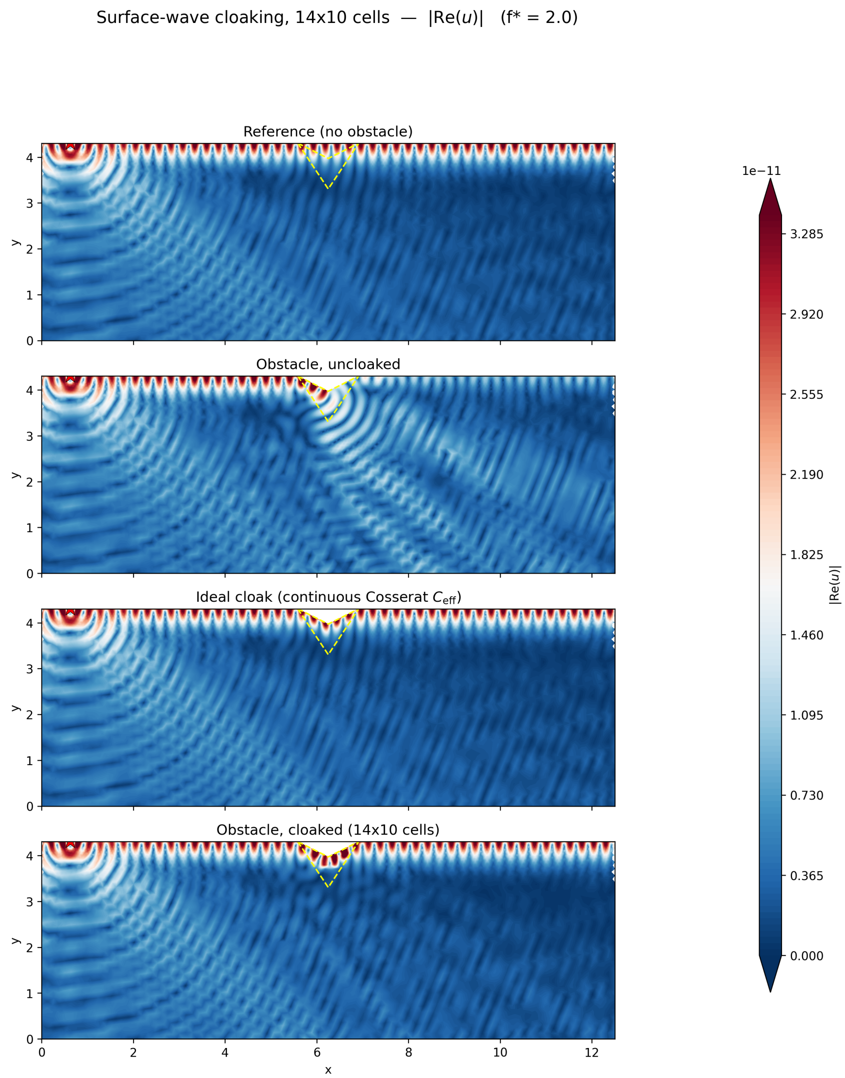


Figure 6: Real-displacement magnitude $|\mathrm{Re}\,\mathbf{u}|$ at $f^\star = 2.0$ (top to bottom): defect-free reference, uncloaked notch, ideal continuous polar cloak $\mathbf{c}_{\mathrm{eff}}$, and the optimised 14×10-cell cloak (46 in-cloak materials, $\eta = 0.990$). The optimised cloak reconstructs the reference wavefield on the transmission side and is visually indistinguishable from the ideal continuous cloak, with the downstream shadow of the uncloaked notch removed. Dashed lines mark the cloak/defect boundary.

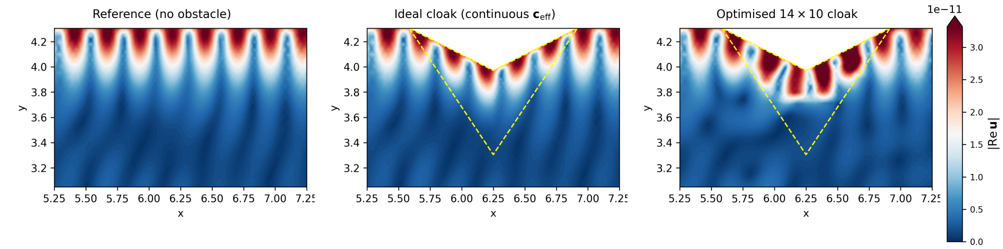


Figure 7: Zoom on the cloak region ($|\mathrm{Re}\,\mathbf{u}|$ at $f^\star = 2.0$, shared colour scale): defect-free reference (left), ideal continuous polar cloak (centre), and optimised 14×10 cloak (right); dashed lines outline the inner (defect) and outer cloak boundaries. All three reproduce the same amplitude envelope, but the optimised cloak leaves its near-surface wavefronts slightly phase-shifted ($\approx 0.18\lambda$, $\approx 65°$) relative to the reference and the ideal cloak — a constant phase offset left unconstrained by the phase-invariant magnitude loss, not a cloaking error.

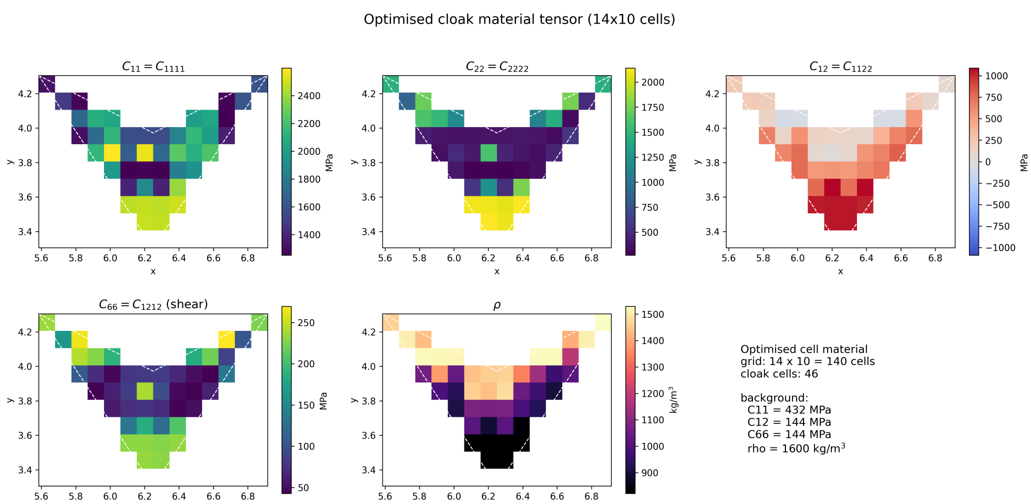


Figure 8: Optimised per-cell material of the 14×10 single-frequency cloak: the four orthotropic Cauchy moduli $C_{11}, C_{22}, C_{12}, C_{66}$ and the density ρ over the 46 cells inside the cloak region (background cells masked). Only these 46 materials are optimised.

single_freq_comparison.png

Figure 9: Frequency response of two independently optimised 50×50 neural-field cloaks, each trained at a single frequency ($f^\star = 2.5$ and $f^\star = 3.0$). The sharp minima at the training frequencies illustrate frequency-specific overfitting.

The ideal sweep furnishes the lower envelope attainable without microstructure constraints; the obstacle sweep is the worst-case upper envelope. Together they bound the optimisation target and let us state per-frequency cloaking quality as a fraction of the available gap (Sec. 5.4). Figure 10 reports both.

reference_sweeps.png

Figure 10: Reference frequency sweeps on the target band: continuous $\mathbf{c}_{\text{eff}}$ cloak (lower envelope) and uncoated defect (upper envelope). Shaded region is the gap closed by optimisation.

5.3. Min–max/mean objective

For a target frequency band $\{f_1, \ldots, f_N\}$ we replace $\mathcal{L}_{\text{cloak}}$ in (**??**) by the convex combination

$$\mathcal{L}_{\text{band}}(\theta) = \alpha \, \max_i \mathcal{L}(f_i;\theta) + (1-\alpha)\,\frac{1}{N}\sum_i \mathcal{L}(f_i;\theta), \tag{22}$$

with α annealed from 1 to 0 during training. The pure min–max regime ($\alpha = 1$) prevents the optimiser from trading performance at one frequency for another; relaxing α towards 0 sharpens the final solution once the loss landscape has been levelled. Figure 11 shows that this schedule produces a flat, low loss over the entire target band $f^\star \in [1.0, 4.0]$.

`multifreq_sweep.png`

Figure 11: Frequency sweep of a single 50×50 neural-field cloak optimised with the multifrequency objective (22).

5.4. Cloaking ratio across the band

The dimensionless cloak ratio $\eta(f) = \langle|\mathbf{u}_{\text{cloak}}(f)|\rangle / \langle|\mathbf{u}_{\text{ref}}(f)|\rangle$ on the free surface beyond the cloak is reported across the target band in Figure 12. The optimised neural-field cloak holds $\eta \approx 1$ throughout the band; the obstacle remains far from unity and the ideal cloak provides a moving target close to but not identically one (the residual gap is the mesh-convergence floor of Sec. **??**). The same ratio is the objective used in the microstructure-constrained pipeline of Sec. 6.

u_ratio_chart.png

Figure 12: Cloaking ratio $\eta(f) = \langle|\mathbf{u}|\rangle/\langle|\mathbf{u}_{\mathrm{ref}}|\rangle$ on the free surface beyond the cloak across the target band, for the obstacle, the continuous $\mathbf{c}_{\mathrm{eff}}$ cloak, and the optimised neural-field cloak. A flat $\eta \approx 1$ is the broadband cloaking ideal.

5.5. Floquet–Bloch dispersion

To verify that the optimised broadband material is physically admissible we compute the Floquet–Bloch dispersion of a vertical strip unit cell cut through the optimised cloak, Bloch-periodic along the surface direction x_1, traction-free at the top and fixed at the bottom, following the construction used for the Rayleigh-wave benchmark [12]. Because the triangular transformation is invariant along x_1, the cell width is unconstrained. Figure 13 reports the lowest surface branches over the reduced Brillouin zone $k_{x_1} \in (0, \pi/L_c]$. The dispersion is real-valued across the target band — the optimised cell supports propagating surface modes and does not rely on locally resonant bandgaps to suppress transmission-side amplitude.

bloch_dispersion.png

Figure 13: Floquet–Bloch dispersion of a vertical strip unit cell through the broadband neural-field cloak, Bloch-periodic along the surface direction. Shaded band is the broadband training target.

5.6. Results

Figure 14 reports the per-frequency cloak loss of the neural-field reparameterisation across the target band. The neural-field run produces a flat, low response across the entire band, reaching a band-averaged metric value of 0.985. Figure 15 shows the optimised (C, ρ) fields; the neural field, unconstrained by a prior and fed high-order Fourier features, produces sharp designs with fine spatial features. Floquet–Bloch analysis of the multifrequency optimum (Sec. 5.5) confirms that the resulting unit cell supports the propagation regime expected at the band centre.

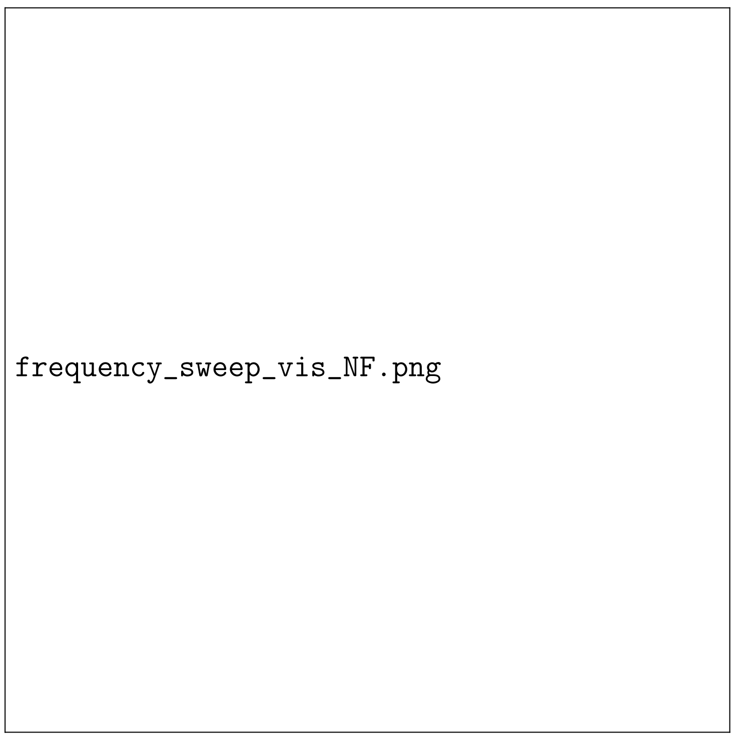


Figure 14: Per-frequency cloak loss of the neural-field reparameterisation across the target band. The broadband training flattens the response with respect to single-frequency optimisation.

6. Microstructure-constrained optimisation

The cloaks of Sec. 5 are tensor-valued: each cell carries a homogenised (C, ρ) that need not correspond to any realisable microstructure. We now material_field_NF.png

Figure 15: Optimised (C, ρ) material fields produced by the neural-field reparameterisation.

constrain the neural field to a manifold of physically realisable two-phase microstructures, following the dataset-generation strategy of Yang et al. [20] for density-based mechanical metamaterials.

6.1. Dataset and homogenisation

A dataset of $N \sim 10^6$ binary 50×50 unit cells is generated by a squared-assembly cellular automaton on a cement/void substrate. Periodic-FEM homogenisation on each unique cell yields effective Lamé parameters $(\lambda_{\text{eff}}, \mu_{\text{eff}})$, density ρ_{eff}, and the full effective stiffness tensor C_{eff}. The construction follows Yang et al. [20] in shape and generative philosophy (binary CA + homogenisation) but is specialised to the elastodynamic cloak target rather than to static stiffness/Poisson design.

6.2. Dataset distribution

Figure 16 shows the marginal and pairwise distributions of the homogenised descriptors $(\lambda_{\text{eff}}, \mu_{\text{eff}}, \rho_{\text{eff}})$ over the $N \sim 10^6$ cell catalogue, with the trajectory traced by the single-frequency optimum overlaid. The dataset covers a band-limited region of the (λ, μ, ρ) space; the cloak-relevant portion of the analytical $\mathbf{c}_{\text{eff}}$ map – shown as the dashed locus – partially exits the dataset envelope, anticipating the projection error quantified in Sec. 6.4.

dataset_distribution.png

Figure 16: Joint distribution of homogenised $(\lambda_{\text{eff}}, \mu_{\text{eff}}, \rho_{\text{eff}})$ over the cellular-automaton dataset. Dashed locus: analytical $\mathbf{c}_{\text{eff}}$ map across the cloak. Coloured points: per-cell optimum of the constrained single-frequency design before snap-to-dataset.

6.3. GMM prior and constrained loss

A Gaussian mixture is fit to the standardised $(\lambda_{\text{eff}}, \mu_{\text{eff}}, \rho_{\text{eff}})$ point cloud, and a flat-top threshold τ at the q-th log-density quantile defines the realisable manifold $\mathcal{M}$. Since the cellular-automaton dataset is intrinsically Cauchy, the neural field here uses the 4CM parameterisation (four free orthotropic Cauchy moduli; Sec. 2.3), summarised by the per-cell isotropic descriptor $(\lambda_c, \mu_c, \rho_c)$. These parameters are optimised through the neural field to minimise the cloak loss together with a manifold-penalty

$$\mathcal{L}_{\text{GMM}} = \sum_c \max\big(0,\, \tau - \log p_{\text{GMM}}(\lambda_c, \mu_c, \rho_c)\big), \tag{23}$$

which pulls every cell into $\mathcal{M}$ without forcing it onto a specific dataset entry. After optimisation each cloak cell is snapped to the dataset entry whose standardised (λ, μ, ρ) is closest in Euclidean distance, and the resulting matched microstructures tile the cloak region for pixel-level validation on a refined mesh ($\sim$1 element per micro-pixel side). Figure 17 summarises the steps.

6.4. Diagnostic experiment

To separate the *matching error* (projecting onto the discrete dataset) from the *pixel-vs.-homogenised error*, we add an intermediate stage: each cloak cell's optimised (λ, μ, ρ) is replaced by its matched dataset (λ, μ, ρ), but the FEM is still run with a single homogenised (C, ρ) per cell rather than the actual pixel microstructure. Table 4 reports the cloak ratio $\langle|u|\rangle/\langle|u_{\text{ref}}|\rangle$ at $f^\star = 2.0$ on a 20×15 macro grid with 100 cloak cells.

Table 4: Microstructure-constrained results: cloak ratio at $f^\star = 2.0$. Mesh refinement does not close the matched-vs-validated gap.

Configuration	$\langle\lvert u\rvert\rangle/\langle\lvert u_{\text{ref}}\rvert\rangle$	mesh refinement
Optimised (homogenised, free λ, μ, ρ)	≈ 1.00	3
Matched (homogenised, dataset λ, μ, ρ)	0.9385	3
Matched (homogenised, dataset λ, μ, ρ)	0.9544	20
Validated (pixel-level, real microstructure)	≈ 0.70	25

pipeline.png

Figure 17: Microstructure-constrained pipeline: CA dataset generation $\rightarrow$ FEM homogenisation $\rightarrow$ GMM prior on (λ, μ, ρ) $\rightarrow$ neural-field optimisation with manifold penalty $\rightarrow$ snap-to-dataset $\rightarrow$ pixel-level FEM validation.

The ∼5% matching gap is benign.. Going from the free optimum (1.00) to the matched homogenised cloak (0.94–0.95) costs about five percentage points. This is the price of projecting onto a discrete dataset and is straightforwardly improved by (i) enlarging the dataset, (ii) reweighting the matching distance so that μ is not effectively ignored, or (iii) replacing nearest-neighbour snap with a generative match.

The ∼25% pixel-vs-homogenised gap is the real problem.. Once the matched entries are inserted as actual microstructures the cloak ratio collapses to ∼0.70 and mesh refinement does not close the gap. The most likely cause is hidden anisotropy: the descriptor (λ, μ) is an isotropic projection of an effective stiffness that is generally orthotropic, and two microstructures with the same (λ, μ) but different principal axes scatter waves very differently. Dynamic-homogenisation effects (Willis couplings [21]) and pixel-level scattering at cement/void interfaces are secondary candidate causes. A disambiguating experiment in which the full anisotropic C tensor of each matched microstructure is recomputed and reinserted will determine whether the gap is removable in principle by storing and optimising over anisotropic moduli, or whether it is a genuinely finite-wavelength effect that requires either dynamic homogenisation or direct optimisation in microstructure space.

6.5. Displacement-field validation

Figure 18 compares the displacement field of the matched homogenised cloak with the pixel-level validation on the refined mesh. The transmission-side amplitude is well reconstructed by the homogenised solve but the pixel-level field carries higher-order features near cell interfaces that the descriptor (λ, μ, ρ) cannot encode. These features are the visual correlate of the ∼25% pixel-vs-homogenised gap quantified in Table 4.

6.6. Microstructure gallery

Figure 19 shows a representative sample of the matched cement/void microstructures tiled across the cloak. Each panel is a 50×50 binary unit cell from the snap-to-dataset matching of Sec. 6.4; the colour outline encodes the position of the host cell in the cloak, illustrating the gradient from strongly sheared, low-density configurations near the defect boundary to near-bulk configurations at the outer boundary.

displacement_field_validation.png

Figure 18: Displacement-field validation at $f^{\star} = 2.0$. Left: reference (defect-free). Centre: matched homogenised cloak. Right: pixel-level validation on the refined mesh. Insets zoom into the cloak interior to highlight scattering at cement/void interfaces.

microstructure_gallery.png

Figure 19: Gallery of matched cement/void microstructures tiling the cloak. Each panel is a 50×50 binary unit cell from the dataset; outline colour encodes the position of the host macro cell in the cloak.

7. Broadband microstructure-constrained design

The single-frequency microstructure pipeline of Sec. 6 extends to the broadband regime by combining the GMM manifold penalty of Eq. (23) with the multifrequency objective (22). We retain the 4CM stiffness parameterisation of Sec. 2.3 – the cellular-automaton dataset is intrinsically Cauchy and is fully covered by the four orthotropic Cauchy moduli – and train a single neural field against

$$\mathcal{L}_{\text{band+micro}}(\theta) = \mathcal{L}_{\text{band}}(\theta) + \lambda_{\text{GMM}}\, \mathcal{L}_{\text{GMM}}(\theta) + \lambda_{\ell_2}\, \mathcal{L}_{\ell_2}(\theta). \tag{24}$$

The manifold penalty is annealed from a small initial value so that the broadband cloak loss can shape the geometry before the GMM constraint tightens around the realisable region.

Snap, validate, sweep.. After optimisation we snap the per-cell parameters to the nearest dataset entry (as in Sec. 6) and run a frequency sweep on the matched homogenised cloak across the same band used in Sec. 5. Figure 20 reports the cloaking ratio $\eta(f)$ across the band, alongside the reference and obstacle sweeps of Sec. 5.2. The broadband micro-constrained cloak retains a roughly flat profile – evidence that the GMM penalty does not collapse the design to a single-frequency optimum – but exhibits the same residual amplitude penalty isolated in Sec. 6.4.

8. Discussion

The three regimes above share a single architectural choice – a coordinate-conditioned neural field on top of a differentiable FEM – and they show, in different forms, the same advantage over hand-crafted symmetrisation strategies. In the single-frequency case the network finds a low-distortion realisable cloak by directly searching four orthotropic Cauchy moduli per cell; the broken minor symmetry belongs to the ideal polar tensor and represents the gap that any realisable Cauchy design (ours included) must approximate. In the multifrequency case the network's implicit smoothness eliminates the spurious cell-to-cell oscillations that plague raw per-cell optimisation and the band objective flattens the loss across a wide target band. In the microstructure-constrained case the same network output is projected onto a realisability manifold, and the resulting diagnostic isolates a homogenisation gap that is hidden in pure tensor-level designs.

multifreq_micro_sweep.png

Figure 20: Frequency sweep of the broadband microstructure-constrained cloak: cloaking ratio $\eta(f) = \langle|\mathbf{u}|\rangle/\langle|\mathbf{u}_{\mathrm{ref}}|\rangle$ for the matched homogenised cloak (solid) and the pixel-level validation on the refined mesh (dashed), against the reference envelopes.

Comparison with prior work.. Wang et al. [11] use the same type of L^2 outside-cloak metric in an elasto-static setting and report a reduction from 17.2% in the voided structure to 4% in the cloaked structure for a circular-void cloak with a database of orthotropic unit cells. Under the analogous metric our continuous $\mathbf{c}_{\text{eff}}$ cloak reduces the outside distortion from $\sim 5\%$ to an extrapolated $\sim 0.75\%$, and the optimised 14×10 neural-field cloak attains $\sim 1.7\%$. The comparison is not strictly apples-to-apples – our setting is frequency-domain elastodynamic (the ideal cloak is polar; our realisable design is orthotropic Cauchy 4CM), theirs is quasi-static and Cauchy – but the order of magnitude is encouraging. The microstructure-constrained results sit between the two regimes and make explicit the (sub)gap that any pure tensor-level optimisation has been silently hiding.

Limitations.. The cell discretisation imposes a piecewise-constant material approximation whose square cells cannot conform exactly to the inclined cloak interfaces; the residual right-boundary distortion of the continuous tensor is dominated by FEM error rather than by an intrinsic limit of the transformation. The microstructure pipeline is currently restricted to a two-phase binary substrate and to an isotropic (λ, μ, ρ) descriptor; the anisotropic generalisation proposed in Sec. 6 is the natural next step.

Future work.. The most direct extension is to stop snapping to a discrete database altogether and to learn a generative model of the realisable manifold that can be queried differentiably during optimisation (e.g. a guided diffusion model in the spirit of Yang et al. [20]). A second extension is to incorporate dynamic homogenisation or Willis-type effective tensors into the microstructure descriptor, which we expect to close most of the remaining homogenisation gap. A third extension is to move from the two-dimensional carpet cloak to three-dimensional configurations, for which the differentiable JAX-FEM pipeline carries over without qualitative change. A fourth extension is to design beyond the four-parameter orthotropic Cauchy class by realising the broken minor symmetry of the ideal polar tensor via Cosserat or polar microstructures, directly closing the gap between the BGM prescription and the realisable design.

9. Conclusion

We have presented a unified, differentiable design pipeline for two-dimensional Rayleigh-wave carpet cloaks in which the per-cell material is parameterised by four free orthotropic Cauchy moduli $(C_{11}, C_{12}, C_{22}, C_{66})$ and a scalar density, searched by a coordinate-conditioned neural field through an end-to-end differentiable JAX-FEM pipeline. This is an alternative to closed-form arithmetic-mean symmetrisation of the ideal polar tensor that directly explores the realisable Cauchy design space; the ideal BGM transformation tensor (polar, ten components) is retained as the mathematical reference, while all optimised designs are Cauchy. The pipeline recovers near-perfect single-frequency cloaking, flattens the cloak loss across a target frequency band when trained with a min–max/mean objective, and can be projected onto a realisability manifold of homogenised binary microstructures to expose a hidden-anisotropy gap that has not, to our knowledge, been isolated before in the elastodynamic regime.

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